Exp 9 · JFET drain and transfer characteristics, and device parameters

Manual pp. 26–28 · LTspice · CO1, CO2 · theory: ECE2101 unit 3 · the simulation reproduces the manual's printed table exactly · the professor flagged these four parameters as an 8-mark question (classmate's report, 2026-10-05)

9 JFET curves → 10 CS amplifier → 11 Oscillator → 12 MOSFET inverter

What you already have: from Exp 1, a reverse-biased pn junction carries no current and has a depletion region with no free carriers. From Exp 5, families of output curves and slopes taken as parameters.

Three terms first:

- **FET (field-effect transistor):** current flows through a **channel**, and a **voltage** on the gate controls the channel's width. It is *voltage-controlled*. A BJT, by contrast, is controlled by base current.
- **JFET terminals: drain** (D, current in), **source** (S, current out), **gate** (G, controls).
- **Unipolar:** only one carrier type carries the current (electrons, in an n-channel JFET). A BJT uses both electrons and holes, which is why it is called *bipolar*.

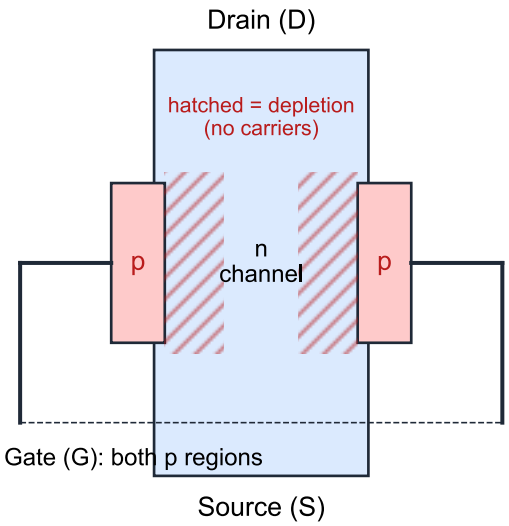
MANUAL EXP 9, AIM · PROFESSOR'S SLIDE

To plot the drain and transfer characteristics of a JFET in CS configuration and determine the device parameters. Viva / 8-mark form: "Define g_m, r_d, g_d and μ, find each from the curves, and show $\mu = g_m r_d$."

GUESS FIRST

Which of the four do you think is read from the *transfer* curve, and which from the *drain* curves?

How a JFET works



n-channel JFET: an n bar between two p gate regions. The gate–channel junctions are kept reverse-biased ($V_{GS} \le 0$).

1. **V_{GS} = 0:** the channel is fully open, and current flows from D to S.
2. **V_{GS} more negative:** the reverse-biased gate junctions' depletion regions widen into the channel, so the channel narrows and I_D falls.
3. **V_{GS} = V_P** (pinch-off voltage, ≈ −3 V here): the depletion regions meet and $I_D \approx 0$. The JFET is off.
4. **Raising V_{DS}** at fixed V_{GS}: at first I_D rises like a resistor (**ohmic region**). Then the channel pinches off near the drain and I_D levels out (**saturation**, or pinch-off region). The amplifier operates here.
5. The gate junction is reverse-biased, so **gate current ≈ 0**: the input resistance is enormous (≫ MΩ). This is the JFET's big advantage.

Shockley's equation (the transfer curve, in saturation):

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$$

- I_{DSS} = drain current with the gate **S**horted to the **S**ource (V_{GS} = 0), in saturation.

The four parameters (the professor's slide)

Parameter	Definition	Held constant	Read from
transconductance g_m	$\Delta I_D/\Delta V_{GS}$	V_{DS}	two drain curves at one V_{DS} (or the transfer curve's slope)
drain resistance r_d	$\Delta V_{DS}/\Delta I_D$	V_{GS}	one drain curve, in saturation
drain conductance g_d	$\Delta I_D/\Delta V_{DS} = 1/r_d$	V_{GS}	same
amplification factor μ	$-\Delta V_{DS}/\Delta V_{GS}$	I_D	**$\mu = g_m \times r_d$**

Why $\mu = g_m r_d$: μ asks how much V_{DS} change balances a V_{GS} change at constant I_D. Write it as a chain of ratios: $(\Delta V_{DS}/\Delta I_D) \times (\Delta I_D/\Delta V_{GS}) = r_d \times g_m$. The minus sign records that a more negative V_{GS} must be balanced by a larger V_{DS}.

Do it

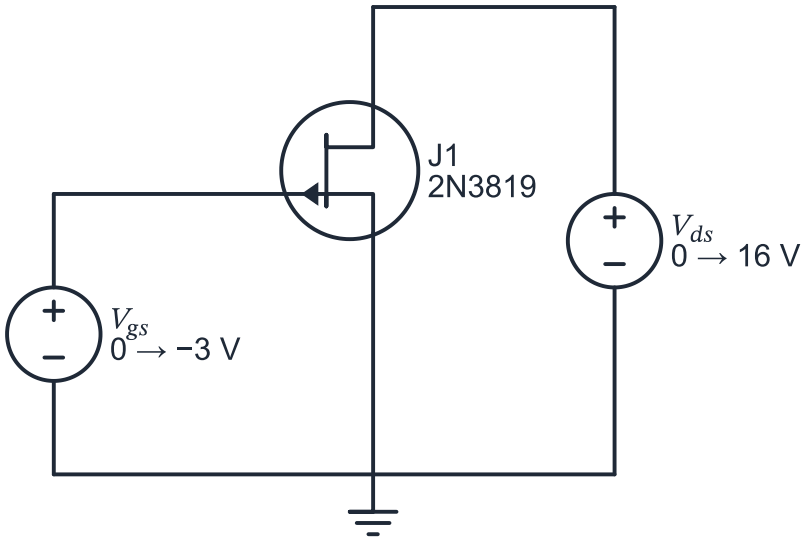
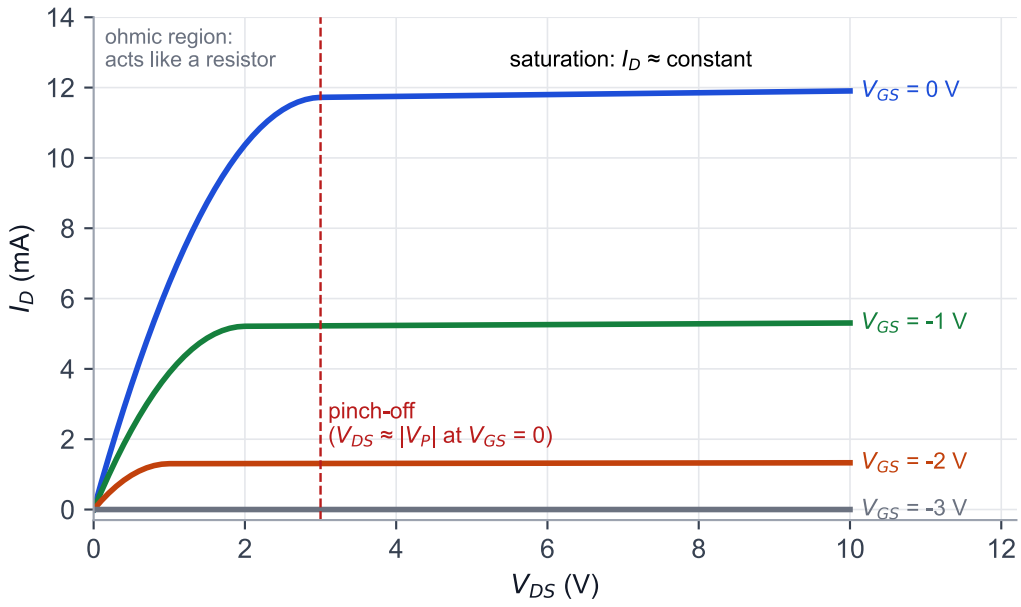


Fig 9.2, redrawn: V_{gs} on the gate, V_{ds} on the drain, source grounded.

DO IT (LTSPICE)

1. Draw the circuit with J1: Pick New JFET → **2N3819** (an n-channel JFET, `njf`). Name the sources `Vgs` and `Vds`.
2. **Drain characteristics:** `.dc Vds 0 16 0.1 Vgs 0 -3 -1`. This gives four curves, for V_{GS} = 0, −1, −2, −3 V. Plot **Id(J1)**.
3. **Transfer characteristic:** set Vds = 8 V, then `.dc Vgs -5 0 0.01`. Plot Id(J1) against V_{gs}.
4. Read I_{DSS} (cursor at V_{GS} = 0) and V_P (where I_D reaches ≈ 0).

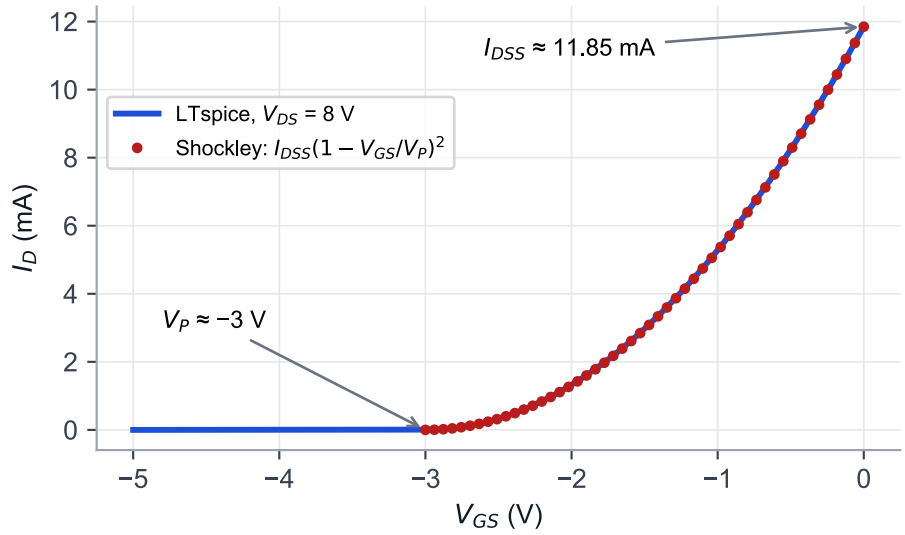
What you should see



Drain characteristics: ohmic rise, pinch-off, then flat saturation. Each curve flattens at $V_{DS} \approx V_{GS} - V_P$.

V_{DS} (V)	0	1	2	3	4	5	6	7	8
I_D (mA), V_{GS} = 0	0	6.45	10.37	11.72	**11.75**	11.77	**11.80**	**11.83**	11.85
I_D (mA), V_{GS} = −1	0	3.89	5.21	5.22	**5.24**	5.25	5.26	5.27	5.28
I_D (mA), V_{GS} = −2	0	1.30	1.31	1.31	1.31	1.32	1.32	1.32	1.32
I_D (mA), V_{GS} = −3	0	0	0	0	0	0	0	0	0

These are the manual's printed numbers (it lists them in amps: 0.01185 A = 11.85 mA).



Transfer characteristic at V_{DS} = 8 V, with Shockley's equation (dots) laid on top: a near-perfect match.

Calculations

- **I_{DSS} = 11.85 mA** (V_{GS} = 0, V_{DS} = 8 V).
- **V_P ≈ −3 V** (I_D reaches 0 at V_{GS} = −3 V).

$$g_m = \frac{\Delta I_D}{\Delta V_{GS}} = \frac{11.75 - 5.24 \text{ mA}}{0 - (-1) \text{ V}} = 6.51 \text{ mA/V} (= 6.51 \text{ mS}) \qquad \text{at } V_{DS} = 4 \text{ V}$$

$$r_d = \frac{\Delta V_{DS}}{\Delta I_D} = \frac{8 - 3 \text{ V}}{11.852 - 11.722 \text{ mA}} = \frac{5 \text{ V}}{0.130 \text{ mA}} = 38.5 \text{ k}\Omega \qquad \text{at } V_{GS} = 0$$

$$g_d = \frac{1}{r_d} = 26 \ \mu\text{S} \qquad \mu = g_m r_d = 6.51 \text{ mS} \times 38.5 \text{ k}\Omega \approx 250$$

Cross-check with Shockley:

- The slope at $V_{GS} = 0$ is $g_{m0} = 2I_{DSS}/|V_P| = 2 \times 11.85/3 = 7.9$ mS.
- The 0 → −1 V chord equals the slope at the midpoint, $V_{GS} = -0.5$ V: $g_m = g_{m0}(1 - 0.5/3) = 6.6$ mS ✓ (6.51 measured).

TRAP: THE MANUAL'S R_D LINE

It writes (7 − 6)/(0.1183 − 0.1180) = 3.3 kΩ. The currents are 0.01183 A and 0.01180 A, so the difference is 0.03 mA, and r_d = 1 V / 0.03 mA = **33 kΩ**. Also, a 1 V span with 4-digit data is coarse: use a wide V_{DS} span (3 → 8 V) inside saturation, which gives 38.5 kΩ.

WRITE IN YOUR RECORD

Result: I_{DSS} = **11.85 mA**, V_P = **−3 V**, g_m = **6.51 mS** (V_{DS} = 4 V), r_d = **38.5 kΩ** (V_{GS} = 0); also g_d = 26 µS and $\mu = g_m r_d \approx 250$.

Theory for the record: an n-channel JFET is an n-type channel between two p-type gate regions. With the gate–source junction reverse-biased, the depletion regions narrow the channel as V_{GS} becomes more negative, so the gate voltage controls the drain current. Gate current is negligible, giving very high input resistance. As V_{DS} increases, I_D first rises linearly (ohmic region), then the channel pinches off and I_D saturates. At $V_{GS} = V_P$ the channel is cut off. In saturation, $I_D = I_{DSS}(1 - V_{GS}/V_P)^2$ (Shockley). Parameters: $g_m = \Delta I_D/\Delta V_{GS}$ at constant V_{DS}; $r_d = \Delta V_{DS}/\Delta I_D$ at constant V_{GS}; $\mu = g_m r_d$.

CHECK 1

(a) Use Shockley to predict I_D at $V_{GS} = -2$ V, and compare with the table. (b) Why is a JFET's gate never forward-biased in normal use? (c) BJT vs JFET: give two differences an examiner would accept.

Answers

Opening question. g_m from the **transfer** curve (or two drain curves at one V_{DS}); r_d and g_d from **one drain curve** in saturation; μ from $g_m \times r_d$.

Check 1. (a) $11.85 \times (1 - 2/3)^2 = 11.85/9 =$ **1.32 mA**; the table gives 1.31–1.32 ✓. (b) Forward bias would make the gate junction conduct. The gate would draw current, the high input resistance would be lost, and the device could be damaged. (c) Any two: a JFET is voltage-controlled, a BJT current-controlled; a JFET is unipolar, a BJT bipolar; a JFET has very high input resistance, a BJT moderate ($h_{ie} \approx$ kΩ); a JFET is less noisy and more thermally stable; a BJT has higher gain (g_m).